

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275569

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Le; Hoang Minh et al.

PREROLL FORMING SYSTEM AND METHOD OF USING THE SAME

Abstract

A system may include an unloading tray configured to receive filled paper preroll cones from a knock box. The unloading tray includes a mold defining apertures, each sized to receive a single cone. A tamping tool has a base and a plurality of rods depending therefrom, each being positioned on the base to align with a corresponding mold aperture. A fold plate includes a plurality of chamfers configured to fold a portion of each cone inwards, each chamfer being positioned to align with a corresponding mold aperture, and defining a hole sized to allow a tamping tool rod to pass therethrough. The tamping tool is positioned with the rods extending through the holes, and receives first and second downward forces, which are applied to the cones, causing the folded cone to close and to pass through the mold apertures, sealing the cones and expelling the cones from the unloading tray.

Inventors: Le; Hoang Minh (Lynnwood, WA), Lee; Jaimie (Lynnwood, WA)

Applicant: Wolfpack Technologies, LLC (Wilmington, DE)

Family ID: 96881315

Appl. No.: 18/593257

Filed: March 01, 2024

Publication Classification

Int. Cl.: A24C5/06 (20060101); A24C5/54 (20060101)

U.S. Cl.:

CPC A24C5/06 (20130101); A24C5/54 (20130101);

Background/Summary

FIELD OF DISCLOSURE

[0001] The present disclosure generally relates to forming of preroll joints, and more specifically to large scale forming of preroll joints for sale at a dispensary.

BACKGROUND

[0002] In the cannabis market, a preroll is a marijuana “joint” that is packed with flower, wrapped in rolling paper, and ready for sale at a dispensary. Prerolls are available at most dispensaries in a variety of strains and quantities. In the highly competitive cannabis market, profit margins can be very slim so it is important that a cannabis operator is able to run an efficient production in order to minimize cost and maximize profit.

[0003] When forming preroll joints, a first operator would, traditionally, use a knock box to fill a number of pre-roll cones and then unload the filled cones from the knock box. Another production technician or operator would then hand tamp and twist the tops of each filled cone to finish the process. This process is labor intensive and can take an experienced technician up to an hour to complete twisting of approximately 100 preroll joints.

[0004] Accordingly, there is a need for production process for preroll joints that can increase output and decrease the level of skill needed for the operators.

BRIEF OVERVIEW

[0005] This brief overview is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This brief overview is not intended to identify key features or essential features of the claimed subject matter. Nor is this brief overview intended to be used to limit the claimed subject matter's scope.

[0006] A preroll filling system is designed to work in collaboration with a knock box. The box may be capable of filling multiple (e.g., 50, 100, 200, etc.) preroll cones at a time.

[0007] The preroll filling system, when used correctly, may allow an operator to produce preroll within 0.1 g of the target weight, and to produce prerolls at a throughput rate of about 2000 prerolls per hour. A team of 4 relatively inexperienced technicians can produce upwards of 16,000 prerolls per day from start to completion.

[0008] An operator may begin by placing an unloading tray beneath a knock box containing a plurality of filled conical rolling papers into a knock box. For example, the knock box may be configured to hold 100 cones. The cones in the knock box may be filled using a fill tray designed to fill the cones to a specified weight desired by the operator (e.g., 1 gram). Once the cones are filled, the operator may unload the cones into the unloading tray of the preroll filling system. The unloading tray allows the knock box to nest into notches on the top of the unloading tray, thus aligning the cones in the knock box with holders in the unloading tray. A trap door disposed on the top of the unloading station may help to prevent the prerolls from inadvertently falling into the incorrect holes, and/or from falling before the knock box tray is nested in the notches of the unloading tray. Once the holders are aligned with the correct holes of the unloading tray, the operator can remove the trap door, allowing the filled cones to fall into the properly matched holes.

[0009] Beneath the trap door, the unloading tray includes a mold designed specifically to match the size of the preroll the operator is working on. The prerolls may fall directly into each respective hole. Once unloaded into the unloading tray, the operator may place the unloading tray on top of a crowning block. The crowning block is a block having one or more holes or cavities disposed therein to align with the holes of the mold of the unloading station. The operator may then carefully align the cones in the mold so that the cones are positioned above the holes of the crowning block.

[0010] The operator may use a tamping tool to tamp the flower within the cones down to the top surface of the mold. The tamping tool may include a plate having a plurality of rods extending

therefrom. Each of the rods may be aligned with a hole in the mold, and may be sized to fit within the cone. The tamping serves two purposes: giving the preroll a proper pack density, and prepping the top of the preroll cone with enough leftover paper to properly fold and close.

[0011] Once tamped, the operator may position a fold plate over the plurality of tamped cones. The fold plate is designed with a plurality of angled chamfers positioned so as to force the tops of the paper cones of the prerolls to close in on themselves. In embodiments, the fold plate is designed with each chamfer including an aperture sized to allow the rods of the tamping tool to extend therethrough. The operator may place the tamping tool onto the fold plate, pushing the tops of the filled preroll cones down, closing the tops of the prerolls. The technician may place the entire stacked pre-roll filling platform underneath a press. The press is designed to apply downward force on the tamping tool, such that the rods of the tamping tool push the prerolls through the holes in the mold of the unloading tray and into the cavities of the crowning block. The press is specially designed with a plate that disperses the energy of the press across the surface area of the tamping tool.

[0012] The press pushes the cones through the silicone mold, effectively closing the tops of the prerolls and providing the preroll with an additional compression allowing for a pack density to promote a slow and even burn. The platform preferably leaves each preroll with a crown on the top, and with the paper tucked inside of the crown to retain the flower inside. Once completed, the technician inspects the preroll for any defects and sends the batch of filled and packed cones to the next technician.

[0013] In some aspects, the techniques described herein relate to a preroll joint forming system, including: an unloading tray configured to receive a plurality of filled paper preroll cones from a knock box into the unloading tray, the unloading tray includes a mold including a plurality of apertures, each aperture sized to receive and retain a single preroll cone, of the plurality of preroll cones; a tamping tool including a base plate and a plurality of rods depending therefrom, wherein each of the plurality of rods is positioned on the base plate to align with a corresponding aperture of the mold, the tamping tool being configured to apply downward pressure to the preroll cones in each aperture of the mold; a fold plate configured to rest on top of the mold and to fold at least a portion of each paper preroll cone inwards, towards a center axis, wherein the fold plate includes a plurality of angled chamfers configured to effect the folding, wherein each of the chamfers is positioned to align with a corresponding aperture of the mold, and wherein each chamfer defines a hole sized to allow a rod of the tamping tool to pass therethrough; wherein the tamping tool is positioned with the plurality of rods extending through the plurality of holes defined in the fold plate, and is configured to receive a first downward force and to apply the first downward force to the preroll cones causing the folded paper cone to close; and wherein the tamping tool is configured to receive a second downward force and to apply the second downward force to the preroll cones, causing the preroll cones to pass through the apertures of the mold, thus sealing the preroll cones and expelling the preroll cones from the unloading tray.

[0014] In some aspects, the techniques described herein relate to a method of forming preroll joints, including: positioning an unloading tray beneath a knock box to receive a plurality of filled paper preroll cones from the knock box into the unloading tray, wherein each of the filled preroll cones is received in a corresponding aperture of a mold; tamping each of the plurality of filled paper preroll cones using a tamping tool including a base plate and a plurality of rods depending therefrom, wherein each of the plurality of rods is positioned on the base plate to align with an aperture of the mold; positioning a chamfered fold plate on top of the plurality of preroll cones to fold at least a portion of each paper preroll cone inwards, towards a center axis, wherein the fold plate includes a plurality of angled chamfers to effect the folding, wherein each of the chamfers is positioned to align with a hole in the mold, and wherein each chamfer defines a hole sized to allow a rod of the tamping tool to pass therethrough; placing the plurality of rods of the tamping tool through the plurality of holes defined in the fold plate to apply a first downward force, thereby

causing the folded paper cone to close; and pressing the tamping tool to apply a second downward force, the second downward force causing the preroll cones to pass through the apertures of the mold, thus sealing the preroll cones and expelling the preroll cones from the unloading tray. [0015] Both the foregoing brief overview and the following detailed description provide examples and are explanatory only. Accordingly, the foregoing brief overview and the following detailed description should not be considered to be restrictive. Further, features or variations may be provided in addition to those set forth herein. For example, embodiments may be directed to various feature combinations and sub-combinations described in the detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various embodiments of the present disclosure. The drawings contain representations of various trademarks and copyrights owned by the Applicant. In addition, the drawings may contain other marks owned by third parties and are being used for illustrative purposes only. All rights to various trademarks and copyrights represented herein, except those belonging to their respective owners, are vested in and the property of the Applicant. The Applicant retains and reserves all rights in its trademarks and copyrights included herein, and grants permission to reproduce the material only in connection with reproduction of the granted patent and for no other purpose.

[0017] Furthermore, the drawings may contain text or captions that may explain certain embodiments of the present disclosure. This text is included for illustrative, non-limiting, explanatory purposes of certain embodiments detailed in the present disclosure. In the drawings:

[0018] FIG. 1 is a diagram showing the parts of the preroll forming system;

[0019] FIG. 2 is a perspective view of an unloading tray of the preroll forming system;

[0020] FIG. 3 is a perspective view of a tamping tool of the preroll forming system;

[0021] FIG. 4 is a perspective view of a fold plate of the preroll forming system; and

[0022] FIG. 5 is a flow chart of a method for using the preroll forming system.

DETAILED DESCRIPTION

[0023] As a preliminary matter, it will readily be understood by one having ordinary skill in the relevant art that the present disclosure has broad utility and application. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the disclosure and may further incorporate only one or a plurality of the above-disclosed features. Furthermore, any embodiment discussed and identified as being “preferred” is considered to be part of a best mode contemplated for carrying out the embodiments of the present disclosure. Other embodiments also may be discussed for additional illustrative purposes in providing a full and enabling disclosure. Moreover, many embodiments, such as adaptations, variations, modifications, and equivalent arrangements, will be implicitly disclosed by the embodiments described herein and fall within the scope of the present disclosure.

[0024] Accordingly, while embodiments are described herein in detail in relation to one or more embodiments, it is to be understood that this disclosure is illustrative and exemplary of the present disclosure and are made merely to provide a full and enabling disclosure. The detailed disclosure herein of one or more embodiments is not intended, nor is to be construed, to limit the scope of patent protection afforded in any claim of a patent issuing here from, which scope is to be defined by the claims and the equivalents thereof. It is not intended that the scope of patent protection be defined by reading into any claim a limitation found herein that does not explicitly appear in the claim itself.

[0025] Thus, for example, any sequence(s) and/or temporal order of steps of various processes or

methods that are described herein are illustrative and not restrictive. Accordingly, it should be understood that, although steps of various processes or methods may be shown and described as being in a sequence or temporal order, the steps of any such processes or methods are not limited to being carried out in any particular sequence or order, absent an indication otherwise. Indeed, the steps in such processes or methods generally may be carried out in various different sequences and orders while still falling within the scope of the present invention. Accordingly, it is intended that the scope of patent protection is to be defined by the issued claim(s) rather than the description set forth herein.

[0026] Additionally, it is important to note that each term used herein refers to that which an ordinary artisan would understand such a term to mean based on the contextual use of the term herein. To the extent that the meaning of a term used herein—as understood by the ordinary artisan based on the contextual use of such term—differs in any way from any particular dictionary definition of such term, it is intended that the meaning of the term as understood by the ordinary artisan should prevail.

[0027] Regarding applicability of 35 U.S.C. § 112, ¶6, no claim element is intended to be read in accordance with this statutory provision unless the explicit phrase “means for” or “step for” is actually used in such claim element, whereupon this statutory provision is intended to apply in the interpretation of such claim element.

[0028] Furthermore, it is important to note that, as used herein, “a” and “an” each generally denotes “at least one,” but does not exclude a plurality unless the contextual use dictates otherwise. When used herein to join a list of items, “or” denotes “at least one of the items,” but does not exclude a plurality of items of the list. Finally, when used herein to join a list of items, “and” denotes “all of the items of the list.”

[0029] The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While many embodiments of the disclosure may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods. Accordingly, the following detailed description does not limit the disclosure. Instead, the proper scope of the disclosure is defined by the appended claims. The present disclosure contains headers. It should be understood that these headers are used as references and are not to be construed as limiting upon the subject matter disclosed under the header.

[0030] The present disclosure includes many aspects and features. Moreover, while many aspects and features relate to, and are described in, the context of a method and/or system for forming cannabis prerolls, embodiments of the present disclosure are not limited to use only in this context.

I. Platform Overview

[0031] This overview is provided to introduce a selection of concepts in a simplified form that are further described below. This overview is not intended to identify key features or essential features of the claimed subject matter. Nor is this overview intended to be used to limit the claimed subject matter's scope.

[0032] A system for creating prerolls may include an unloading tray positioned beneath a knock box containing a plurality of filled conical rolling papers into a knock box. As one example, the knock box may be configured to hold 100 cones, though more or fewer ones are also possible. The system may optionally include a fill tray designed to fill the cones to a specified weight desired by the operator (e.g., 1 gram).

[0033] The unloading tray allows the knock box to nest into notches on the top of the unloading tray, thus aligning the cones in the knock box with holders in the unloading tray. A trap door disposed on the top of the unloading station may help to prevent the prerolls from inadvertently falling into the incorrect holes, and/or from falling before the knock box tray is nested in the

notches of the unloading tray. Once the holders are aligned with the correct holes of the unloading tray, the trap door may be removed, allowing the filled cones to fall into the properly matched holes. Beneath the trap door, the unloading tray includes a mold designed specifically to match the size of the preroll the operator is working on. The prerolls may fall directly into each respective hole.

[0034] The system may further include a crowning block, having one or more holes or cavities disposed therein. The holes or cavities of the crowning block may align with the holes of the mold of the unloading station.

[0035] A tamping tool may be designed to tamp the flower within the cones down to the top surface of the mold. The tamping tool may include a plate having a plurality of rods extending therefrom. Each of the rods may be aligned with a hole in the mold, and may be sized to fit within the cone. The tamping serves two purposes: giving the preroll a proper pack density, and prepping the top of the preroll cone with enough leftover paper to properly fold and close.

[0036] The system may include a fold plate designed to cause the conical rolling papers to fold, so as to form the preroll. The fold plate may include a plurality of angled chamfers positioned so as to force the tops of the paper cones of the prerolls to close in on themselves. In embodiments, the fold plate is designed with each chamfer including an aperture sized to allow the rods of the tamping tool to extend therethrough.

[0037] The tamping tool may further be used in conjunction with the fold plate, pushing the tops of the filled preroll cones down, closing the tops of the prerolls. A press may be designed to apply downward force on the tamping tool, such that the rods of the tamping tool push the prerolls through the holes in the mold of the unloading tray and into the cavities of the crowning block. The press may include a plate that disperses the energy of the press across the surface area of the tamping tool, facilitating an even tamping of all prerolls.

[0038] The press may be configured to push the cones through the silicone mold, effectively closing the tops of the prerolls and providing the preroll with an additional compression allowing for a pack density to promote a slow and even burn. The system preferably leaves each preroll with a crown on the top, and with the paper tucked inside of the crown to retain the flower inside.

[0039] Embodiments of the present disclosure may comprise methods and systems comprising, but not limited to, at least one of the following: [0040] A. An Unloading Tray; [0041] B. A Tamping Tool; [0042] C. A Fold Plate;

[0043] In some embodiments, the present disclosure may provide an additional set of modules or components. The additional set of modules may comprise, but not be limited to: [0044] D. A Crowning Block; [0045] E. A Knock Box; [0046] F. A Fill Tray; and [0047] G. A Press.

[0048] Details with regard to each module are provided below. Although modules are disclosed with specific functionality, it should be understood that functionality may be shared between modules, with some functions split between modules, while other functions duplicated by the modules. Furthermore, the name of each module should not be construed as limiting upon the functionality of the module. Moreover, each component disclosed within each module can be considered independently, without the context of the other components within the same module or different modules. Each component may contain functionality defined in other portions of this specification. Each component disclosed for one module may be mixed with the functionality of other modules. In the present disclosure, each component can be claimed on its own and/or interchangeably with other components of other modules.

[0049] The following depicts an example of a method of a plurality of methods that may be performed by at least one of the aforementioned modules, or components thereof. Various hardware components may be used at the various stages of the operations disclosed with reference to each module. For example, although methods may be described to be performed by a single computing device, it should be understood that, in some embodiments, different operations may be performed by different networked elements in operative communication with the computing device. For

example, at least one computing device *00 may be employed in the performance of some or all of the stages disclosed with regard to the methods. Similarly, an apparatus may be employed in the performance of some or all of the stages of the methods. As such, the apparatus may comprise at least those architectural components as found in computing device *00.

[0050] Furthermore, although the stages of the following example method are disclosed in a particular order, it should be understood that the order is disclosed for illustrative purposes only. Stages may be combined, separated, reordered, and various intermediary stages may exist. Accordingly, it should be understood that the various stages, in various embodiments, may be performed in orders that differ from the ones disclosed below. Moreover, various stages may be added or removed without altering or departing from the fundamental scope of the depicted methods and systems disclosed herein.

[0051] Consistent with embodiments of the present disclosure, a method may be performed by at least one of the modules disclosed herein. The method may be embodied as, for example, but not limited to, computer instructions which, when executed, perform the method. The method may comprise the following stages: [0052] Positioning an unloading tray beneath a knock box to receive a plurality of filled paper preroll cones from the knock box into the unloading tray, wherein each of the filled preroll cones is received in a corresponding aperture of a mold; [0053] Tamping each of the plurality of filled paper preroll cones using a tamping tool comprising a base plate and a plurality of rods depending therefrom, wherein each of the plurality of rods is positioned on the base plate to align with an aperture of the mold; [0054] Positioning a chamfered fold plate on top of the plurality of preroll cones to fold at least a portion of each paper preroll cone inwards, towards a center axis, wherein the fold plate comprises a plurality of angled chamfers to effect the folding, wherein each of the chamfers is positioned to align with a hole in the mold, and wherein each chamfer defines a hole sized to allow a rod of the tamping tool to pass therethrough; [0055] Placing the plurality of rods of the tamping tool through the plurality of holes defined in the fold plate to apply a first downward force, thereby causing the folded paper cone to close; and [0056] Pressing the tamping tool to apply a second downward force, the second downward force causing the preroll cones to pass through the apertures of the mold, thus sealing the preroll cones and expelling the preroll cones from the unloading tray.

[0057] Both the foregoing overview and the following detailed description provide examples and are explanatory only. Accordingly, the foregoing overview and the following detailed description should not be considered to be restrictive. Further, features or variations may be provided in addition to those set forth herein. For example, embodiments may be directed to various feature combinations and sub-combinations described in the detailed description.

II. Platform Configuration

[0058] As shown in FIGS. **1-4**, embodiments of the present disclosure provide a system **100** comprised of a set of components including, but not limited to:

A. Unloading Tray

[0059] As shown in FIGS. **1-2**, the system **100** may include an unloading tray **102**. The unloading tray **102** may include one or more side walls **110**, one or more knock box supports **120**, and a mold **130**.

[0060] The one or more side walls **110** provide a support for the unloading tray **102**. Each of the one or more side walls is preferably formed from a sturdy, rigid material such as, but not limited to, wood, metal, plastic (e.g., polyethylene, polyurethane, etc.) and/or the like. The one or more side walls **110** may support the mold **130**.

[0061] The knock box supports **120** preferably extend upwards from peripheral edges of the unloading tray **102**. The height of the knock box supports may be selected such that a knock box can be positioned above the unloading tray. In embodiments, each of the knock box supports **120** may include a notch **122** to facilitate alignment of the knock box with the unloading tray **102**.

[0062] The mold **130** is preferably formed from an elastic materials, such as silicone. In

embodiments, the mold **130** may include a top surface **132** and a plurality of apertures **134**. For example, each of the apertures **134** may have a generally circular shape at the top surface **132**. In some embodiments, each of the apertures **134** may taper or narrow. Each aperture **134** is preferably sized to receive a conical rolling paper, and so that a predetermined amount of the rolling paper extends beyond the top surface **132** when the rolling paper is seated within the aperture **134**. The apertures **134** defined by the mold may be sized to prevent the contents of the cavity from passing therethrough without application of a significant external force. When force is applied to the contents of the cavity, the aperture may allow the contents (e.g., the conical rolling paper and the contents thereof) to pass through the aperture without tearing the rolling paper.

[0063] As shown in FIG. 2, the mold **130** may include 100 apertures **134**. However, the mold **130** may include more or fewer apertures **134** without departing from the scope of the invention. In particular, the number and arrangement of the apertures **134** may be dictated by the number and position or pattern of the conical rolling papers held by the knock box. That is, the number and arrangement of the apertures **134** may be selected so as to align with the knock box when the knock box is positioned within the notches **122**.

[0064] In some embodiments, the unloading tray **102** may include a moveable cover or door **140**. The door **140** may be movable between a closed position and an open position. In the closed position, the door **140** may block access to the apertures **134** from above. In the open position, the door is moved to permit access the apertures **134** from above. The door may be placed in the closed position to prevent the rolling paper from falling into the holes before the knock box is aligned with the unloading tray **102**. thereafter, the door may be moved to the open position (or removed from the assembly entirely) to permit the rolling papers to pass from the knock box to the apertures **134** of the unloading tray **102**.

B. Tamping Tool

[0065] As shown in FIGS. 1 and 3, the system **100** may include a tamping tool **300**. The tamping tool **300** may include a base **302** and a plurality of rods **304** depending therefrom. The base **302** is preferably formed from a rigid material, such as wood, metal, plastic, etc. Each rod **304** is preferably from a relatively rigid and incompressible material. In embodiments, it may be beneficial for the rods **304** to be formed from or coated with a material that discourages sticking, such as polytetrafluoroethylene (PTFE). In some embodiments, the tamping tool may be used to tamp or compact the contents of the conical rolling papers to increase the packing density of the contents. Additionally or alternatively, the tamping tool may be used to press the conical rolling papers (and contents thereof) through the mold **130** to seal the preroll joints.

[0066] In embodiments, each rod **304** may have a diameter that is approximately equal to the diameter of a preroll joint. As a non-limiting example, the rod **304** may have a diameter of approximately 8 mm. The length of each rod may be selected to allow for tamping and extrusion of the preroll joints through the mold. As an example, each rod may have a length of approximately 2 inches.

[0067] The base **302** of the tamping tool **300** is preferably sized to approximately match the size of the mold **130**. Moreover, the plurality of rods **304** on the tamping tool **300** are preferably disposed to align with the apertures **134** of the mold **130**. That is, as shown in FIG. 3, the tamping tool includes **100** rods **304**. However, the tamping tool **300** may include more or fewer rods **304** without departing from the scope of the invention. In particular, the number and arrangement or pattern of the rods **304** may be dictated by the number and position or layout of the apertures **134**. That is, the number and arrangement of the rods **304** may be selected so as to align with the apertures **134** when the tamping tool **300** is operatively positioned above and/or within the apertures **134**.

C. Fold Plate

[0068] As shown in FIGS. 1 and 4 the system **100** may include a fold plate **400**. The fold plate **400** may be a plate that rests atop the mold **130** to help fold and close the tops of the filled prerolls. In embodiments the fold plate **400** may be sized to match the top surface **132** of the mold **130**. The

fold plate **400** may include a lower surface **402** and an opposing upper surface **404**. The lower surface **402** of the fold plate **400** may be configured to about the top surface **132** of the mold **130**. The fold plate **400** may define a plurality of apertures **406** disposed in the fold plate.

[0069] Each of the apertures **406** may be substantially circular at the lower surface **402** of the fold plate **400**. In embodiment, each aperture **406** may include a chamfered wall **408**. The chamfered wall **408** is angled to cause the top edges of the conical rolling papers to fold inward, towards a central axis of the rolling paper, when the fold plate **400** is disposed on the top surface **132** of the mold **130**. For example, the chamfered wall **408** may have an angle of approximately 45 degrees, with respect to a plane defined by the lower surface **402**. In other embodiments, the chamfered wall **408** may define an angle between 30 and 60 degrees with respect to the plane defined by the lower surface **402**. At the upper surface **404**, each aperture **406** may have a diameter sized to receive a rod **304** of the tamping tool **300**. For example, where the rod **304** has a diameter of 8 mm, the aperture **406** may have a diameter that is 8.5 mm at the upper surface **404**.

[0070] The fold plate **400** is preferably sized to approximately match the size of the mold **130**. Moreover, the plurality of apertures **406** on the fold plate **400** are preferably disposed to align with the apertures **134** of the mold **130**. That is, as shown in FIG. 4, fold plate **400** includes 100 apertures **406**. However, the fold plate **400** may include more or fewer apertures **406** without departing from the scope of the invention. In particular, the number and arrangement or pattern of the fold plate apertures **406** may be dictated by the number and position or layout of the mold apertures **134**. That is, the number and arrangement of the fold plate apertures **406** may be selected so as to align with the mold apertures **134** when the fold plate **400** is operatively positioned above the mold **130**.

D. Crowning Block

[0071] As shown in FIG. 1, the system **100** may optionally include a crowning block **150**. The crowning block **150** may provide separation between the unloading tray **102** and a work bench or table (not shown). In embodiments, the crowning block **150** may include one or more cavities (not shown) within the block to collect preroll joints expelled from the unloading tray **102** through the apertures **134** of the mold **130**.

E. Knock Box

[0072] In some embodiments, the system **100** may optionally include a knock box (not shown). The knock box may be used in conjunction with a fill tray to fill the conical rolling papers. The knock box may hold a plurality of conical rolling papers for filling. The number and arrangement of the conical rolling papers held by the knock box may dictate the number and arrangement of apertures **134** in the mold **130**.

F. Fill Tray

[0073] In some embodiments, the system **100** may optionally include a fill tray (not shown). The fill tray may be used to fill the conical rolling papers in the knock box. In embodiments, the fill tray may fill each rolling paper with its contents (e.g., cannabis flower) to a level (e.g., a weight) established by the operator. In some embodiments, the fill tray may facilitate filling the conical rolling papers to the level specified by the operator within a tolerance of 0.1 g.

G. Press

[0074] In some embodiments, as shown in FIG. 1 the system **100** may optionally include a press **200**. The press **200** may be, for example, a pneumatic press, hydraulic press mechanical press, or any other press designed to provide a downward force to the tamping tool **300**. The press **200** may be configured to distribute the downward force substantially evenly across the tamping tool **300**. The downward force provided by the press **200** may be sufficient to provide additional tamping and to extrude the filled and closed conical rolling papers through the mold **130** and out of the unloading tray **102**. The press **200** helps to ensure consistent application of force from tray to tray, as well as even application of force across the tamping tool **300**, when compared to force applied by a human, for example.

III. Platform Operation

[0075] Embodiments of the present disclosure provide a hardware system that operates the aforementioned modules in accordance with the methods. The following depicts an example of at least one method of a plurality of methods that may be performed by at least one of the aforementioned modules. Various hardware components may be used at the various stages of operations disclosed with reference to each module.

[0076] Furthermore, although the stages of the following example method are disclosed in a particular order, it should be understood that the order is disclosed for illustrative purposes only. Stages may be combined, separated, reordered, and various intermediary stages may exist. Accordingly, it should be understood that the various stages, in various embodiments, may be performed in arrangements that differ from the ones described below. Moreover, various stages may be added or removed from the without altering or departing from the fundamental scope of the depicted methods and systems disclosed herein.

A. Master Method

[0077] Consistent with embodiments of the present disclosure, a method may be performed by at least one of the aforementioned modules. The method may comprise the following stages:

[0078] An operator may begin by placing an unloading tray beneath a knock box containing a plurality of filled conical rolling papers into a knock box. For example, the knock box may be configured to hold 100 cones. The cones in the knock box may be filled using a fill tray designed to fill the cones to a specified weight desired by the operator (e.g., 1 gram). Once the cones are filled, the operator may unload the cones into the unloading tray of the preroll filling system. The unloading tray allows the knock box to nest into notches on the top of the unloading tray, thus aligning the cones in the knock box with holders in the unloading tray. A trap door disposed on the top of the unloading station may help to prevent the prerolls from inadvertently falling into the incorrect holes, and/or from falling before the knock box tray is nested in the notches of the unloading tray. Once the holders are aligned with the correct holes of the unloading tray, the operator can remove the trap door, allowing the filled cones to fall into the properly matched holes.

[0079] Beneath the trap door, the unloading tray includes a mold designed specifically to match the size of the preroll the operator is working on. The prerolls may fall directly into each respective hole. Once unloaded into the unloading tray, the operator may place the unloading tray on top of a crowning block. The crowning block is a block having one or more holes or cavities disposed therein to align with the holes of the mold of the unloading station. The operator may then carefully align the cones in the mold so that the cones are positioned above the holes of the crowning block.

[0080] The operator may use a tamping tool to tamp the flower within the cones down to the top surface of the mold. The tamping tool may include a plate having a plurality of rods extending therefrom. Each of the rods may be aligned with a hole in the mold, and may be sized to fit within the cone. The tamping serves two purposes: giving the preroll a proper pack density, and prepping the top of the preroll cone with enough leftover paper to properly fold and close.

[0081] Once tamped, the operator may position a fold plate over the plurality of tamped cones. The fold plate is designed with a plurality of angled chamfers positioned so as to force the tops of the paper cones of the prerolls to close in on themselves. In embodiments, the fold plate is designed with each chamfer including an aperture sized to allow the rods of the tamping tool to extend therethrough. The operator may place the tamping tool onto the fold plate, pushing the tops of the filled preroll cones down, closing the tops of the prerolls. The technician may place the entire stacked pre-roll filling platform underneath a press. The press is designed to apply downward force on the tamping tool, such that the rods of the tamping tool push the prerolls through the holes in the mold of the unloading tray and into the cavities of the crowning block. The press is specially designed with a plate that disperses the energy of the press across the surface area of the tamping tool.

[0082] The press pushes the cones through the silicone mold, effectively closing the tops of the

prerolls and providing the preroll with an additional compression allowing for a pack density to promote a slow and even burn. The platform preferably leaves each preroll with a crown on the top, and with the paper tucked inside of the crown to retain the flower inside. Once completed, the technician inspects the preroll for any defects and sends the batch of filled and packed cones to the next technician.

[0083] FIG. 5 is a flow chart setting forth the general stages involved in a method **500** consistent with an embodiment of the disclosure for use of the preroll forming system **100**.

[0084] Method **500** may begin at stage **510** where an operator may position an unloading tray beneath a knock box to receive a plurality of paper preroll cones. The number of paper cones included in the knock box may vary (e.g., the knock box may hold, 50, 100, 200, or more paper preroll cones). In some embodiments, the knock box may be for use with commercial prerolling operations. In some embodiments, one or more (e.g., each) of the paper preroll cones may be filled with a substance (e.g., cannabis flower) using a filling tray and the knock box.

[0085] In some embodiments, the unloading tray may include one or more notches or detents to facilitate aligning the unloading tray with the knock box. Once the unloading tray is aligned with the knock box, the plurality of paper cones from the knock box may be dropped into the unloading tray. For example, the unloading tray may include a trap door or cover that prevents access to the unloading tray from above when the cover is in a closed position. The cover may be positioned in the closed position while aligning the unloading tray with the knock box. Then, responsive to correctly aligning the unloading tray and the knock box, the cover may be moved to an open position (or removed completely, allowing the conical paper cones to fall into the unloading tray).

[0086] In some embodiments, each of the plurality of conical papers may be received in a corresponding aperture of a mold in the unloading tray. Each aperture may be sized such that, when received in the aperture, at least a portion of the conical preroll paper extends upward above a top surface of the mold.

[0087] From stage **510**, where an unloading tray is positioned beneath a knock box to receive a plurality of paper preroll cones, method **500** may advance to stage **520** where each of the plurality of filled preroll cones may be tamped. In embodiments, each of the plurality of prerolls may be tamped substantially simultaneously. For example, the operator may employ a tamping tool having a base plate and a plurality of rods depending therefrom. The plurality of rods may be positioned on the base plate to align with the apertures in the mold.

[0088] To tamp or compact the prerolls, the tamping tool may be positioned above the unloading tray and aligned such that each rod contacts a corresponding one of the prerolls. Thereafter, a downward force may be applied to the tamping tool to tamp the prerolls. Preferably, the tamping may pack or compact the preroll such that the contents of the preroll does not extend about the top surface of the mold, leaving only paper extending above the top surface of the mold. In embodiments, tamping may serve at least two purposes: giving the preroll a proper pack density, and prepping the top of the preroll cone with enough leftover paper to properly fold and close.

[0089] Once the prerolls are tamped in stage **520**, method **500** may continue to stage **530** where the tops of the prerolls may be folded. For example, folding the prerolls may include positioning a chamfered fold plate on top of the plurality of preroll cones to fold at least a portion of each paper preroll cone inward on itself, towards a center axis. The folding may help to seal the preroll, restricting contact between the contents of the preroll and air.

[0090] To facilitate the folding, the fold plate may include a plurality of angled chamfers. The plurality of chamfers may be arranged so as to align with the plurality of apertures in the mold, with each of the chamfers is positioned to align with a hole in the mold.

[0091] After folding the prerolls in stage **530**, method **500** may proceed to stage **540** where computing device *00 may close the plurality of prerolls. In particular, to close the prerolls the operator may use the tamping tool to secure the folds formed in stage **530**. For example, each chamfer of the fold plate may define a hole sized to allow a rod of the tamping tool to pass

therethrough. The operator may place the tamping tool onto the fold plate and through the holes in the chamfers, pushing the tops of the filled, tamped, and folded preroll cones down, thus closing the tops of the prerolls.

[0092] After securing the folds in stage **540**, method **500** may proceed to stage **550** where the prerolls may be pressed out of the unloading tray. To press the prerolls out of the unloading tray, a downward force may be exerted on the tamping tool. The downward force may put pressure on the prerolls, causing each preroll to pass through a corresponding aperture of the mold. As the preroll passes through the mold aperture, the mold may seal the preroll cone and expelling the preroll cones from the unloading tray. Alternatively or additionally, pressing the prerolls through the mold may provide the preroll with an additional compression, allowing for a pack density that promotes a slow and even burn.

[0093] In some embodiments, the downward force may be generated directly by an operator pressing downward on the tamping tool. Alternatively or additionally, the downward force may be generated by a device, such as (but not limited to) a hydraulic press, a pneumatic press, a mechanical press, or any other press. The use of a device to press the tamping tool helps to ensure application of consistent pressure from batch to batch, as well as helping to ensure even pressure is applied across the tamping tool at each preroll in the batch.

[0094] Optionally, the prerolls expelled from the unloading tray may be collected in a crowning block. Alternatively, the prerolls may be expelled onto a workbench or other surface supporting the unloading tray.

[0095] In some embodiments, the expelled prerolls may be inspected for defects. After completing the inspection, at least a subset of the prerolls (e.g., those that pass inspection without defects) may be sent on to another operator or technician (e.g., for packaging). Another subset of the prerolls (e.g., those with identified defects) may be directed to another workflow. For example, the defective prerolls may be sent to quality control personnel to determine what caused the defect, and/or may be disassembled to recover the contents of the preroll.

IV. Claims

[0096] While the specification includes examples, the disclosure's scope is indicated by the following claims. Furthermore, while the specification has been described in language specific to structural features and/or methodological acts, the claims are not limited to the features or acts described above. Rather, the specific features and acts described above are disclosed as examples for embodiments of the disclosure.

[0097] Insofar as the description above and the accompanying drawing disclose any additional subject matter that is not within the scope of the claims below, the disclosures are not dedicated to the public and the right to file one or more applications to claims such additional disclosures is reserved.

Claims

1. A preroll joint forming system, comprising: an unloading tray configured to receive a plurality of filled paper preroll cones from a knock box into the unloading tray, the unloading tray includes a mold comprising a plurality of apertures, each aperture sized to receive and retain a single preroll cone, of the plurality of preroll cones; a tamping tool comprising a base plate and a plurality of rods depending therefrom, wherein each of the plurality of rods is positioned on the base plate to align with a corresponding aperture of the mold, the tamping tool being configured to apply downward pressure to the preroll cones in each aperture of the mold; a fold plate configured to rest on top of the mold and to fold at least a portion of each paper preroll cone inwards, towards a center axis, wherein the fold plate comprises a plurality of angled chamfers configured to effect the folding, wherein each of the chamfers is positioned to align with a corresponding aperture of the mold, and wherein each chamfer defines a hole sized to allow a rod of the tamping tool to pass therethrough;

wherein the tamping tool is positioned with the plurality of rods extending through the plurality of holes defined in the fold plate, and is configured to receive a first downward force and to apply the first downward force to the preroll cones causing the folded paper cone to close, and wherein the tamping tool is configured to receive a second downward force and to apply the second downward force to the preroll cones, causing the preroll cones to pass through the apertures of the mold, thus sealing the preroll cones and expelling the preroll cones from the unloading tray.

2. The system of claim 1, further comprising a crowning block configured to support the unloading tray, wherein the crowning block comprises one or more cavities defined therein, and wherein each of the one or more cavities is disposed so as to collect the preroll cones expelled from the unloading tray.

3. The system of claim 1, further comprising a pneumatic press configured to apply the second downward force to the tamping tool.

4. The system of claim 1, wherein the mold is formed from silicone.

5. The system of claim 1, wherein the preroll cones received in the unloading tray are filled to a specific weight, and wherein a diameter of each aperture of the mold is determined based on the specific weight of the preroll cones.

6. The system of claim 1, wherein the unloading tray further comprises a cover movable between: a closed position and an open position, wherein the cover prevents the filled preroll cones from passing from the knock box to the unloading tray, and an open position, wherein the preroll cones are permitted to pass from the knock box to the unloading tray.

7. The system of claim 1, wherein the rods of the tamping tool are formed from polytetrafluoroethylene.

8. The system of claim 1, wherein the apertures defined in the mold are disposed in a pattern based on a layout of the knock box.

9. The system of claim 1, wherein the chamfers of the fold plate comprise an angled wall disposed at an angle of about 45 degrees, relative to a plane defined by the fold plate.

10. The system of claim 1, wherein the preroll cones are filled with cannabis flower, and where the first downward force and second downward force are configured to compact the cannabis flower to a density that promotes a slow and even burn.

11. A method of forming preroll joints, comprising: positioning an unloading tray beneath a knock box to receive a plurality of filled paper preroll cones from the knock box into the unloading tray, wherein each of the filled preroll cones is received in a corresponding aperture of a mold; tamping each of the plurality of filled paper preroll cones using a tamping tool comprising a base plate and a plurality of rods depending therefrom, wherein each of the plurality of rods is positioned on the base plate to align with an aperture of the mold; positioning a chamfered fold plate on top of the plurality of preroll cones to fold at least a portion of each paper preroll cone inwards, towards a center axis, wherein the fold plate comprises a plurality of angled chamfers to effect the folding, wherein each of the chamfers is positioned to align with a hole in the mold, and wherein each chamfer defines a hole sized to allow a rod of the tamping tool to pass therethrough; placing the plurality of rods of the tamping tool through the plurality of holes defined in the fold plate to apply a first downward force, thereby causing the folded paper cone to close; and pressing the tamping tool to apply a second downward force, the second downward force causing the preroll cones to pass through the apertures of the mold, thus sealing the preroll cones and expelling the preroll cones from the unloading tray.

12. The method of claim 11, wherein the second downward force is applied by a pneumatic press.

13. The method of claim 12, wherein the second downward force expels the sealed preroll cones into a cavity defined in a crowning block disposed under the unloading tray.

14. The method of claim 11, further comprising inspecting the sealed preroll cones for defect.

15. The method of claim 11, wherein the mold defines a plurality of apertures disposed in a pattern based on a layout of the knock box.

16. The method of claim 11, wherein the paper preroll cones are filled to a specified weight, and wherein a size of each aperture in the mold is selected based on the specified weight.

17. The method of claim 11, wherein the rods of the tamping tool are formed from polytetrafluoroethylene.

18. The method of claim 11, wherein the chamfers of the fold plate comprise an angled wall disposed at an angle of about 45 degrees, relative to a plane defined by the fold plate.

19. The method of claim 11, wherein the unloading tray further comprises a cover movable between: a closed position and an open position, wherein the cover prevents the filled preroll cones from passing from the knock box to the unloading tray, and an open position, wherein the preroll cones are permitted to pass from the knock box to the unloading tray.

20. The method of claim 11, wherein the preroll cones are filled with cannabis flower, and where the first downward force and second downward force compact the cannabis flower to a density that promotes a slow and even burn.
